

DPCN Assignment 1: Opinion Network Formation from an Ordinal Attitude Questionnaire

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Abstract

We construct and analyse an opinion network from a 60-item ordinal attitude questionnaire answered by 96 respondents across four thematic blocks (technology, education, society, environment). The work has three parts. First, we sanitise the data while preserving the distinction between two different missingness mechanisms: structural blanks left when a respondent abandons the survey, and scattered “No Comments” (NC) responses inside otherwise complete questionnaires. NC is not recoded as Neutral and NC users are not discarded; missing responses are instead recovered by question-wise regularised proportional-odds ordinal logistic regression, validated by artificial masking under both missingness patterns. Second, we show that the response distribution is severely ceilinged—79.8% of valid answers fall on the agree side and the grand mean is $+1.14$ on a $[-2, +2]$ scale—so that raw similarity measures how agreeable a respondent is rather than what they believe. Row-centring each respondent removes this component and reduces the number of item pairs correlated at $|r| \geq 0.30$ from 327 to 42. Third, we build correlation networks over the sanitised matrix and run Deffuant–Weisbuch bounded-confidence dynamics on a respondent network. We show analytically and by controlled experiment that the widely reported cluster plateau at small confidence thresholds is an arithmetic artefact of the five-point response grid, while the difference between the sparse network and a well-mixed population is a genuine effect of topology.

1 Introduction

Likert questionnaires are a natural setting for studying how attitudes in different domains relate. Each item is answered independently, yet responses depend on one another, and a network makes those dependencies explicit: items become vertices and statistical association becomes weighted edges.

Turning a questionnaire into a network is not mechanical. Three decisions determine whether the graph

carries signal or merely re-describes the response format. How missing data is handled: abandonment and deliberate non-response are different mechanisms [10], and assigning Neutral to a withheld answer invents data. Whether response style is removed: when respondents agree with almost everything, every pair of items correlates through shared agreeableness rather than content [9]. What the network is compared against: a correlation-derived graph has transitivity built into its construction, so clustering and modularity appear even in structureless data. This report addresses all three, and reports the alternatives that failed alongside the pipeline that survived.

Team and code. Team Edge Runners. All code, figures and data are at https://github.com/shreyash-lohare/dpcn_a1_opinion_network and the whole analysis regenerates from a fresh clone with `python run_all.py`.

2 Data and Preparation

The questionnaire contains 60 items in four blocks of 15 — technology (T01–T15), education (E01–E15), society and ethics (S01–S15) and environment (V01–V15) — answered by 96 respondents. Five ordered categories are encoded symmetrically as $SD = -2$ through $SA = +2$, which represents the ordering without asserting equal spacing; a sixth option, No Comments (NC), is treated as missing. Internal consistency varies sharply by block: Cronbach’s α [3] is 0.87 for environment and 0.82 for society but only 0.65 for technology and 0.58 for education.

2.1 The response ceiling

The most consequential property of this dataset is its ceiling. Figure 1 shows 79.8% of valid answers on the agree side, 43.1% Strongly Agree, and only 7.0% disagreeing. The grand mean is $+1.14$ on the $[-2, +2]$ scale. Two consequences drive every later decision: item means barely discriminate, so the signal is in which statements attract disagreement; and two respondents look similar under any raw similarity measure simply by both being agreeable.

The ceiling is not uniform. Environment ($\mu = 4.39$ on the 1–5 scale, 89.5% agree) and society ($\mu = 4.28$)

are near consensus, while technology ($\mu = 3.99$) and education ($\mu = 3.91$, 13.8% disagree) retain real disagreement — and those two blocks also have the weakest internal consistency, consistent with their items tapping several distinct opinions rather than one settled disposition.

Ranked by disagreement, the twelve most contested items are seven education, four technology, one society, none environment. Compulsory attendance (E03, mean 2.06, 70.1% disagreeing) is the only statement the cohort actively rejects; written examinations (E02, 2.84) is next. The pattern is specific rather than oppositional — the same cohort agrees at 94.3% that curricula should be updated more often (E11). What is contested is assessment and teaching format, not the institution. Technology behaves likewise: teaching AI (T04) and AI as a learning aid (T02) score above 79%, while AI-assisted diagnosis (T08, 3.08) and autonomous vehicles (T09, 3.28) sit lowest. Since robots in hazardous work (T07) scores 82.4%, the objection is to delegating consequential judgment, not to automation.

2.2 Missing data and how it was recovered

Of 96 records, 85 are complete, 5 are entirely empty, and 6 stop partway. The five empty records are dropped, leaving 91 respondents; none is dropped merely for being incomplete. That leaves 205 structural blanks plus 37 NC cells, **242** in total. Dropout is monotone — a truncated tail, not scattered gaps — so later blocks rest on a slightly smaller base.

NC is not Neutral. Recoding a withheld answer as Neutral asserts a response nobody gave. A two-way additive model $\hat{x}_{iq} = \bar{\mu} + a_i + b_q$ fitted on the clean responses predicts 3.92 at the 37 NC cells against 3 for Neutral, so recoding would impose a systematic 0.92-point downward shift. NC use also shows no item- or respondent-level structure (item-uniformity $p = 0.27$; NC users’ mean 4.01 vs 4.15, $p = 0.27$), so there is no basis for assigning it any fixed category.

Recovery by ordinal regression. For target item j the other 59 items form the predictor vector, and we fit one regularised proportional-odds ordinal logistic regression [7] per item, with latent score $z_i = \beta_0 + \beta^\top \mathbf{x}_{i,-j}$ and ordered thresholds $\theta_1 < \dots < \theta_4$; the prediction is the maximum-probability category. This respects the ordering of the scale without assuming equal intervals. Predictors that are themselves missing are filled with the training-fold column median, computed per fold to prevent leakage, and a ridge penalty controls the 59-dimensional fit.

Table 1: Artificial-masking validation. The model beats a modal baseline under both mechanisms, by a wider margin where more co-answered items survive.

Masking	Ordinal model		Mode baseline	
	Acc.	MAE	Acc.	MAE
Scattered (NC-like)	.539	.534	.486	.626
Suffix (structural)	.498	.612	.482	.655

Does it work? The true values are unavailable, so we validate by artificial masking under both mechanisms: scattered masking hides 37 observed cells inside complete respondents (40 trials), suffix masking hides a contiguous tail after a simulated stopping point (15 respondents per trial, 15 trials). Because the target is ordinal, MAE matters more than exact accuracy: being one category out is not the same error as being four.

One caution carries forward. Comparing four NC treatments at $\tau = 0.30$, the correlation matrices agree at $r > 0.94$ while their community assignments agree at AMI as low as 0.337 (dropping NC users, which discards 17 respondents to remove 37 cells). Stability of a correlation matrix is not evidence that a partition of it is stable — a warning that Section 4.2 makes quantitative.

2.3 Row-centring

Imputation yields a complete 91×60 matrix, but not one that can be correlated directly: with a grand mean of +1.14, a respondent who agrees with everything raises all 60 entries together, which reads as association between every pair of items at once. We replace each entry by its deviation from that respondent’s own mean,

$$x'_{i,j} = x_{i,j} - \bar{x}_{i\cdot}, \quad \bar{x}_{i\cdot} = \frac{1}{60} \sum_k x_{i,k}. \quad (1)$$

Centring is applied strictly after imputation and strictly before correlation. Note that rank correlation does not remove this bias automatically: Spearman is invariant to a monotone transform of a column, not to an additive shift that differs by row. Figure 4 measures how much it matters.

2.4 What the network is

Vertices are the 60 questionnaire items. Edges are signed correlations between items, computed on the row-centred matrix and retained with their sign. Edges are weighted by default and binarised only where a metric requires it (path lengths, triad counts). Items were chosen over respondents deliberately: a respondent network must be built by a neighbourhood rule

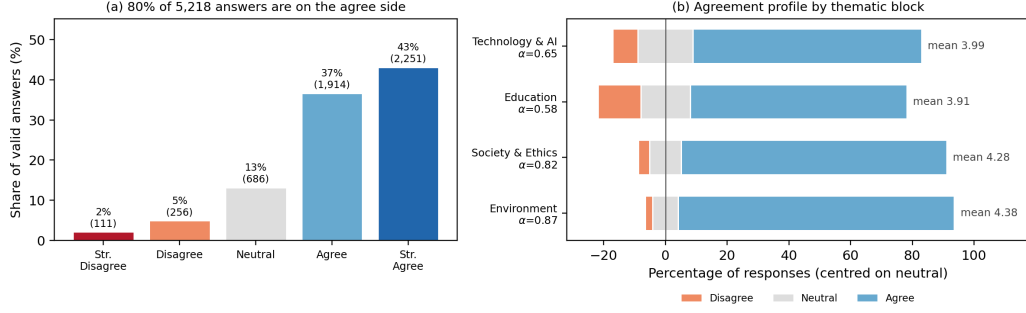


Figure 1: The response ceiling. (a) All 5,218 valid answers: nearly four in five are on the agree side. (b) Agreement profile per block with block mean and Cronbach’s α . Environment and society are near consensus; education carries the most disagreement and the weakest consistency.

such as k -nearest-neighbours, and Section 4.2 shows that rule manufactures its own community structure, whereas an item network is thresholded directly on a measured quantity and every vertex carries an interpretable label. Signs are kept because a negative edge — two questions dividing the cohort in opposite directions — is a stronger statement than mere association, and because the structural-balance analysis is defined on the signs of triads. A second, respondent-level network appears only in Section 3.11, for the dynamics experiment.

3 Analysis and Visualisations

This section reports the twelve analyses. Each states what the figure shows, what was found, and what follows from it. Sections still owned by other team members are marked as placeholders.

A note on respondent counts. Several reference values circulated during planning assume a pipeline variant that drops respondents with more than 12 missing items, leaving 86. This report does not use that filter: all 91 non-empty respondents are retained and their missing cells recovered by ordinal regression (Section 2.2). Figures produced here therefore report $n = 91$, and small deviations from those reference numbers are expected.

3.1 Graph 1: Missingness map and response distribution

Figure 2 shows every cell of the raw 96×60 matrix as missing or present, respondents sorted by missing count, beside the distribution of per-respondent means.

Findings. The 542 missing cells (505 structural blanks plus 37 NC) are extremely unevenly distributed: five entirely empty respondents account for 300 of them, and the remaining 242 spread thinly across the other

91 — a rate of 4.4%. Below the cutoff the map is almost clean, with two visible tails from respondents who stopped early. The right panel locates the grand mean at 1.14, with only one respondent below zero.

Implication. This justifies the retention policy directly: all but five respondents answered at least 80% of items, so there is no population of badly-incomplete respondents whose removal would simplify anything, and a count-based exclusion rule would discard usable respondents to remove very few cells. The right panel is the first direct evidence of the acquiescence problem that row-centring corrects.

3.2 Graph 2: Item variance ranking

Figure 3 ranks all 60 items by standard deviation on the uncentred matrix. Centring is deliberately not applied here: it changes each item’s variance and would answer a different question (spread after removing each respondent’s baseline) than the one asked (how divided the cohort is on the statement as posed).

Findings. Item SD ranges from 0.50 (E15, near-unanimous) to 1.18 (E04, online learning). The five most contested are E04, E03, E02, T08, T07. By block mean SD: technology 0.92, education 0.88, society 0.79, environment 0.72.

Implication. Technology has the highest average variance, but the sharpest disagreements are educational — three of the five most divided items are E02–E04, all about assessment and teaching format — and environment contributes none of the top ten. This is a property of the instrument, not of preprocessing: the questionnaire contains one settled domain, one moderately settled, and two live ones. Consensus items carry little information for a correlation network, since an item everyone endorses cannot covary with anything, so the signal is necessarily concentrated in the contested tail.

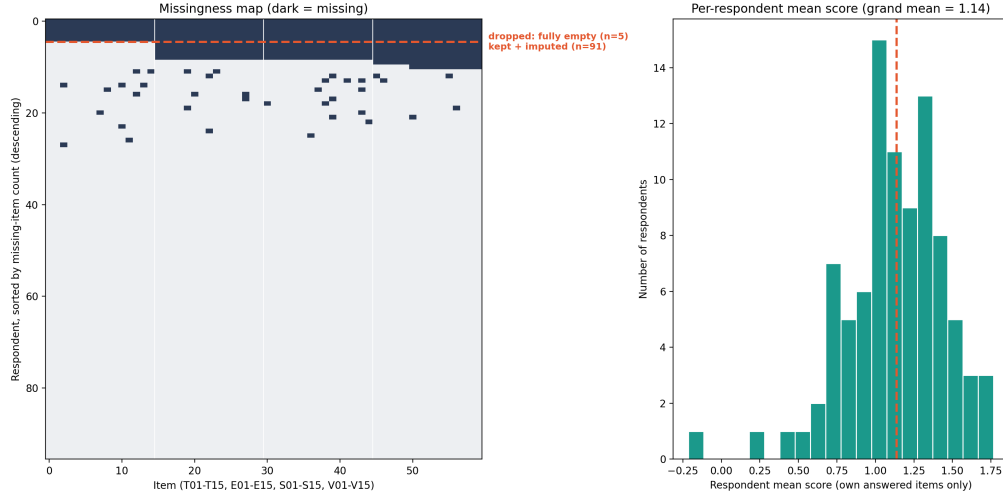


Figure 2: Left: missingness map over all 96 raw respondents, sorted by missing-item count. The dashed line marks the exclusion rule — only the five entirely empty records are dropped, and the remaining 91 have their 242 missing cells imputed. Right: per-respondent mean over answered items, grand mean 1.14.

3.3 Graph 3: The effect of row-centring

Figure 4 overlays all $\binom{60}{2} = 1,770$ pairwise item correlations before and after row-centring, on identical bin edges.

Findings. The raw distribution is displaced positive (mean and median both 0.151), with almost no pair appreciably negative. After centring it sits on zero (mean -0.004 , median -0.008) with positive skew 0.221 leaving a heavier right tail of genuinely co-varying pairs. At $|r| \geq 0.30$ the surviving pairs fall from **327** to **42** — a reduction of 87%.

Implication. This converts the centring argument from an assertion into a measurement: seven eighths of the apparent association in the raw matrix is shared response style. Built on raw scores the network would have been dense, almost entirely positive, and would have supported confident but spurious conclusions. The emergence of negative correlations after centring — invisible in the raw distribution — is also what makes the signed analysis of Graph 11 possible at all.

3.4 Graph 4: Real versus shuffled correlations

Every analysis below runs on the same 91-respondent, ordinal-imputed matrix, so no part of this report silently uses a different sample.

Figure 5 compares the 1,770 observed item–item correlations against a null that destroys inter-item association while preserving every item’s marginal exactly. Each of 200 replicates permutes every column independently and re-runs the full pipeline — row-centre, then correlate — on the shuffled matrix. This detail is not cosmetic: one permutation applied to all

columns, or permuting rows, merely relabels respondents and leaves the correlation structure numerically untouched.

Findings. Correlations from $n = 91$ respondents scatter about zero with SD $1/\sqrt{n-3} = 0.107$ even when nothing is related. The permuted null reproduces this at **0.106**, against **0.139** observed. Real structure exists but is modest, and lives in the tails:

$ r >$	0.15	0.20	0.25	0.30
Observed edges	502	266	124	51
Expected by chance	286	107	32	7
Chance fraction	57%	40%	26%	14%

Implication. At the conventional cutoff $|r| > 0.15$ more than half the surviving edges are what 1,770 simultaneous estimates from 91 people produce by luck. A network drawn there is mostly noise, and no downstream community detection can separate the two after the fact. This makes threshold selection a measured trade-off rather than a convention, and supplies the chance curve that Graph 7 is read against.

3.5 Graph 5: Eigenvalue spectrum against Marchenko–Pastur

Graph 4 establishes that a noise floor exists but not how many genuine dimensions sit above it. PCA cannot settle this: PC1 explains only 8.3% of variance, equally consistent with weak structure and with none. Marchenko–Pastur theory [5] answers it exactly — for a correlation matrix from T observations of N pure-noise variables the eigenvalues fall inside an analytically known band, and anything above its upper edge

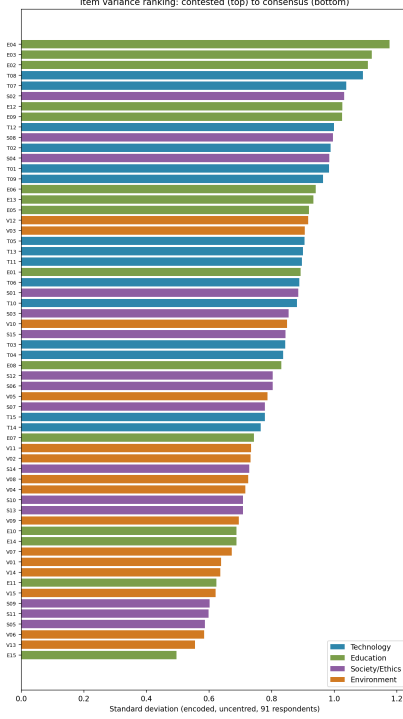


Figure 3: All 60 items by standard deviation on the uncentred matrix, coloured by block. Contested items concentrate in education and technology; environment clusters at the consensus end.

carries more structure than T observations could produce by chance.

Findings. With $T = 91$, $N = 60$, $Q = 1.517$ and $\lambda_+ = (1 + 1/\sqrt{Q})^2 = \mathbf{3.283}$. The bulk of the spectrum tracks the theoretical density closely and exactly four eigenvalues clear the band — 4.38, 4.06, 3.55, 3.35 — carrying **25.6%** of variance (Figure 6). The count is comfortable rather than marginal: the fourth clears by 2.0% while the fifth, at 2.86, sits well inside the band.

Reading the loadings, the four modes are (i) environmental and social commitment against technology governance and assessment tradition, (ii) developmental optimism against regulatory caution, (iii) AI-enabled education and industry against environmental stewardship, and (iv) systemic technological action against individual and community obligation. Modes (i) and (iii) split cleanly along topic blocks; (ii) and (iv) cut across them and are offered as readings of the loadings rather than established constructs.

Implication. Three quarters of this correlation matrix is sampling noise. That disciplines everything downstream: the network is rebuilt from the four significant modes alone, with the diagonal restored to unity, so later community and balance analyses operate on the part of the matrix that 91 respondents can actually support.

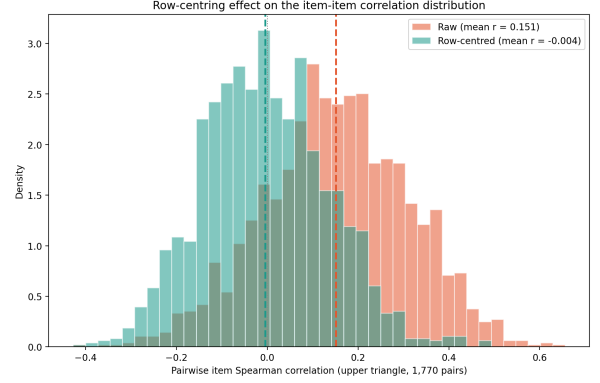


Figure 4: All 1,770 pairwise item correlations before and after row-centring, shared bin edges. Centring moves the mean from 0.151 to -0.004 and removes 87% of pairs exceeding $|r| \geq 0.30$.

3.6 Graph 6: Reordered correlation heatmap

Figure 7 presents the MP-denoised correlation matrix reordered by the community assignments detected in Graph 10. A diverging colour map (centred at zero) distinguishes positive from negative correlations. The marginal colour strips represent the predefined thematic blocks (Technology, Education, Society/Ethics, Environment).

Findings and Implication. The block-diagonal structure of the reordered matrix confirms that the community detection successfully groups densely connected items. However, inspecting the marginal strips reveals that these data-driven communities cut directly across the survey’s predefined thematic blocks. The formal comparison yields a Normalised Mutual Information (NMI) of 0.201, confirming that respondents’ actual opinion structures do not partition neatly into the four hypothesised domains. **Findings and Implication.** The block-diagonal structure of the reordered matrix confirms that the community detection successfully groups densely connected items. However, inspecting the marginal strips reveals that these data-driven communities cut directly across the survey’s predefined thematic blocks. The formal comparison yields a Normalised Mutual Information (NMI) of 0.205, confirming that respondents’ actual opinion structures do not partition neatly into the four hypothesised domains.

3.7 Graph 7: Threshold and percolation sweep

Reporting a network at one threshold invites the objection that the threshold was chosen to produce the result. Figure 8 answers it by rebuilding the signed graph at all 46 cutoffs from 0.05 to 0.50 and tracking edge count, giant component, fragmentation and Lou-

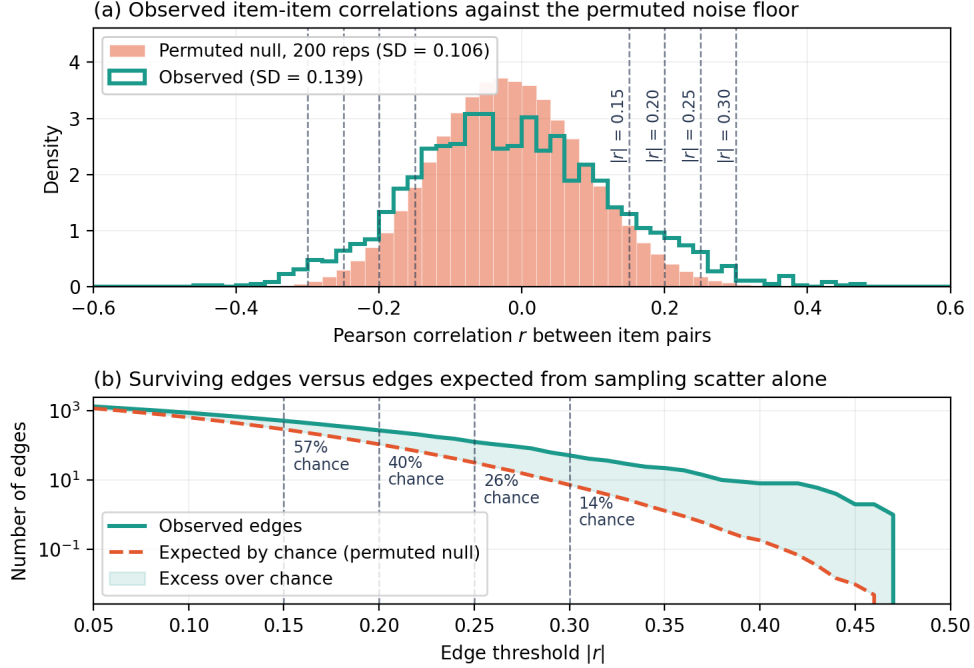


Figure 5: (a) Observed correlations against a 200-replicate permutation null that re-runs the whole pipeline on independently shuffled columns; SD 0.139 against 0.106, the excess confined to the tails. (b) Surviving edges against edges expected from sampling scatter alone (log scale); shaded band is the excess over chance.

vain modularity [1, 8] across the range, so the chosen value is visibly a point on a curve. The sweep also absorbs the giant-component analysis entirely.

Findings. The two constraints pull opposite ways: raising the threshold lowers the chance fraction from Graph 4, but the graph eventually shatters — by $|r| > 0.27$ the giant component holds under half the items. We therefore state the rule explicitly: the highest cut-off at which the giant component still spans more than two thirds of the items and mean degree remains above 4, so community and path metrics stay meaningful. Since spectral denoising has already removed the sampling noise the chance column measures, raising the cutoff further would penalise the same noise twice. This selects $|r| > \mathbf{0.22}$: **131** edges (68 positive, 63 negative), mean degree 4.37, a giant component of 50 of 60 items (83%), modularity $Q = 0.472$. At the next step, $|r| > 0.23$, mean degree falls to 3.47 and the rule stops being satisfied.

Two independent controls agree there despite sharing no machinery. The permutation null implies the raw graph’s 205 edges contain about 67 chance edges, leaving roughly 138 real; spectral denoising, which never sees the null, retains 131 — a 5% difference.

Implication. Panel (d) is the cautionary one. Modularity climbs steadily to 0.72 as the threshold rises, but by then the graph is 43 components with 38 isolated items. Quoting a single high modularity without

the sweep beside it would report fragmentation as community structure — the exact error the null models of Section 4.2 are designed to catch.

3.8 Graph 9: Topic-block connectivity

The questionnaire arrives with its own partition — fifteen items each for technology, education, society and environment — which is an editorial claim about what belongs together. Whether it is also a statistical one should be asked before any community result is interpreted against it. We collapse the 60×60 correlation matrix into a 4×4 block matrix, averaging the 105 distinct within-block pairs on the diagonal (self-correlations excluded: keeping the unit diagonal would manufacture the very block structure being tested). Significance comes from a block-label permutation — the matrix is held fixed and the T/E/S/V labels reshuffled over 5,000 seeded draws — which needs no threshold and is immune to any constant offset in the matrix.

Findings. Row-centring is what makes the design visible at all. On the raw matrix (Figure 9, left) all sixteen cells are positive (+0.041 to +0.377): acquiescence makes every topic look connected to every other. After centring, the sign pattern separates cleanly — every diagonal cell positive, every off-diagonal negative. Within-block mean $r = +0.0746$ against between-block -0.0282 gives a gap of $\mathbf{+0.1027}$ against a null

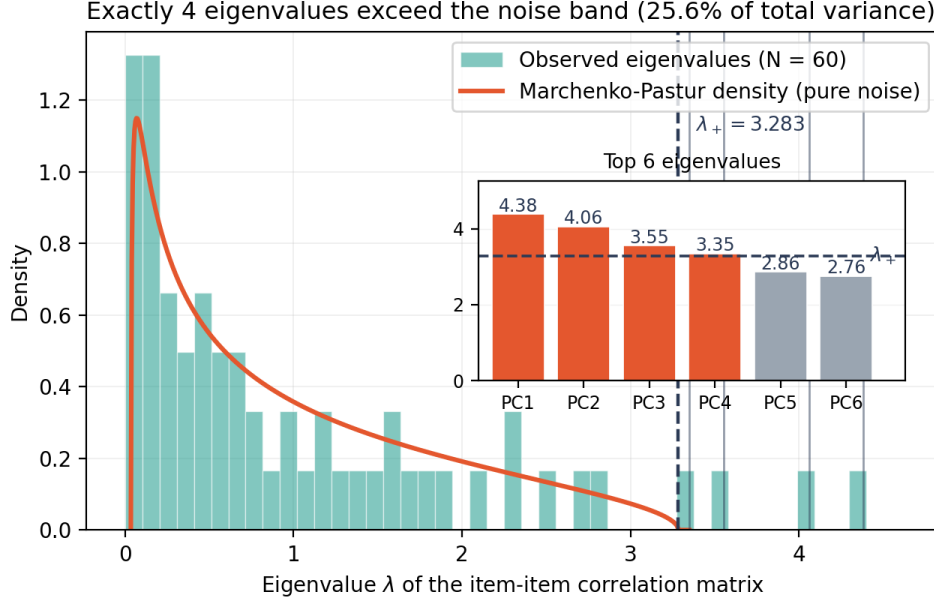


Figure 6: Eigenvalue spectrum of the row-centred item correlation matrix against the Marchenko–Pastur density for pure noise of the same shape ($T = 91$, $N = 60$). Exactly four eigenvalues exceed $\lambda_+ = 3.283$, carrying 25.6% of variance. Inset: a clear gap between the fourth (3.35) and fifth (2.86).

of -0.00003 ± 0.0077 : $\mathbf{z} = \mathbf{13.4}$, with no permutation in 5,000 reaching the observed value ($p < 0.0002$).

The blocks are not equally coherent: environment $+0.136$, technology $+0.066$, education $+0.054$, society/ethics $+0.042$. Environment is more than three times as internally consistent as ethics. S–V is the one seam, at -0.004 the only pair not clearly separated and the strongest off-diagonal cell in the raw matrix ($+0.256$).

Implication. The thematic design is statistically real, so a later community that aligns with a block is recovering something the instrument built in, while one that cuts across blocks is the more informative object — which is exactly what Graph 10 finds (NMI 0.205). The uneven diagonal sharpens this: environmental attitudes travel as a package, whereas ethics items are close to independent, so respondents hold ethical positions issue by issue rather than as a doctrine. One caveat: the negative off-diagonals do not mean topics oppose one another. Row-centring forces each row to sum to zero, pushing the grand mean of all 1,770 correlations to -0.0038 ; the values should be read relative to the diagonal, not as opposition.

3.9 Graph 10: Communities against null models

Figure 11 compares the community structure of the item network at $|r| > 0.22$ against three nulls: degree-preserving rewiring [6], an Erdős–Rényi graph, and a column-permutation null that scrambles responses in-

dependently per item. Figure 10 compares three detection algorithms on the same graph.

Which algorithm. We began with Louvain [1], which found 15 communities at $Q = 0.472$. Louvain has a documented flaw: its greedy aggregation phase can leave communities internally disconnected, capping partition quality [11]. Leiden adds a refinement phase that guarantees well-connected communities and found a strictly better partition, $Q = 0.491$, again at $k = 15$. Girvan–Newman [8], a divisive method that repeatedly removes the highest-betweenness edge, peaks at $Q = 0.447$ — also at $k = 15$ — and then declines monotonically. That all three strategies converge on the same 15 communities makes the partition size a robust feature rather than an artefact of one optimiser, so Leiden is used for the primary analysis.

And then the nulls disagree. Leiden’s $Q = 0.491$ clears the Erdős–Rényi null (0.420 ± 0.016 , $z = 4.46$) and the rewiring null (0.333 ± 0.013 , $z = 12.13$) decisively. Against the column-permutation null it does not merely fail, it falls below it: 0.629 ± 0.044 , $z = -3.14$, $p = 1.000$. Scrambled data pushed through the identical pipeline produces more modular graphs than the real data does. This is the same verdict the respondent network returns in Section 4.2, reached independently on a different network: the modularity is manufactured by thresholding a correlation matrix estimated from 91 respondents, not carried by the opinions.

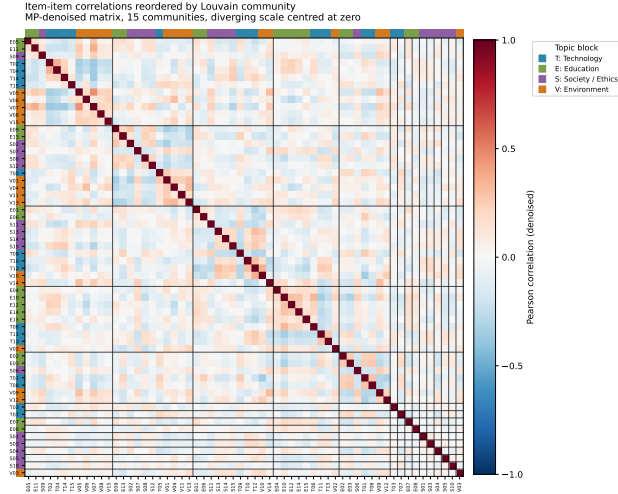


Figure 7: Item-item correlations (MP-denoised), reordered by Leiden community. The 15 communities are separated by black lines, with topic blocks marked in the margins. The communities cut across the predefined topic blocks, showing that respondents’ association patterns do not follow the survey’s own design.

3.10 Graph 11: Structural balance

Figure 12 tracks the fraction of balanced triads in the signed item network as the edge-inclusion threshold $|r|$ tightens, comparing it against a sign-permutation null model [2].

Findings and Implication. The observed fraction of balanced triads is remarkably high and strengthens monotonically as noise is filtered out. At $|r| > 0.15$, 99.1% of triads are balanced. At $|r| > 0.20$ this rises to 99.5%, and for $|r| \geq 0.22$, all triads (100%) are balanced. The sign-permutation null stays flat at roughly 50% across all thresholds. That the balance fraction approaches unity as weak edges are removed constitutes strong evidence that the network’s local topology reflects genuine structural balance, rather than random assortment.

3.11 Graph 12: Opinion dynamics

This is the only analysis involving an actual process; everything above describes a fixed dataset. Opinions belong to respondents, so the vertices here are respondents. From the row-centred matrix we connect each respondent to their $k = 4$ most similar peers, symmetrised by union — mutual k NN fragments this data, and disconnected components cannot merge at any confidence threshold, which would inflate the cluster count for reasons unrelated to the dynamics. The result is 91 nodes, 298 edges, one connected component.

Opinions $x_i \in [0, 1]$ are seeded from E03 (compul-

sory attendance, the most contested item), rescaled by $s(x) = (x + 2)/4$. At each step a random edge is drawn and the pair moves to its midpoint if they are close enough [4]:

$$|x_u - x_v| < \varepsilon \implies x_u, x_v \leftarrow \frac{1}{2}(x_u + x_v). \quad (2)$$

We sweep ε from 0.05 to 0.60, 20 seeded repetitions each.

Findings. The cluster count is exactly 5.00 ± 0.00 for every $\varepsilon \leq 0.25$, then falls to 4.45 at 0.30, 3.35 at 0.40 and 2.00 at 0.50 (Figure 13).

The plateau is an artefact — provably. Under $s(x) = (x+2)/4$ the five categories map to $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, so any two distinct opinions differ by at least $\frac{1}{4}$. For $\varepsilon \leq 0.25$ the merge condition is therefore never satisfied: the initial state is a fixed point and the final count equals the number of occupied categories, on any graph whatsoever. Three checks confirm it empirically. The plateau has zero variance across 20 seeded runs, which no emergent outcome would. A complete graph — a maximally different topology — returns the identical 5.00 ± 0.00 . And jittering the seeds by ± 0.125 , destroying the grid while preserving the distribution, removes the plateau entirely ($35.95 \rightarrow 8.95$ clusters across $\varepsilon = 0.05 \dots 0.25$). The five clusters are the original response categories failing to merge, not opinion groups forming, and the break at $\varepsilon = 0.30$ is arithmetic rather than a discovered threshold.

What is not an artefact. Above $\varepsilon = 0.25$ the comparison is meaningful. An identically seeded complete graph — the same dynamics with topology removed — collapses to about 2 clusters across $\varepsilon = 0.30\text{--}0.45$, while the sparse k NN network sustains 3.15–4.45. Network structure sustains opinion diversity that a well-mixed population loses at the same tolerance: interacting only with nearest neighbours lets locally cohesive groups stabilise before global consensus propagates. This is the one result here that is neither a scale artefact nor a restatement of the input distribution.

4 Results and Discussion

4.1 What the data supports

Superficial consensus, genuine local structure. The headline is not polarisation. Agreement runs at 79.8% across every block and PC1 explains about 8% of variance, so there are no two opposed camps. There is a broad agreeable consensus with real disagreement concentrated in identifiable places.

Disagreement is domain-specific, not dispositional. Environment and society are settled; technology and education are not. Within education the cohort is precise about what it rejects — compulsory attendance

Threshold sweep on the MP-denoised item network

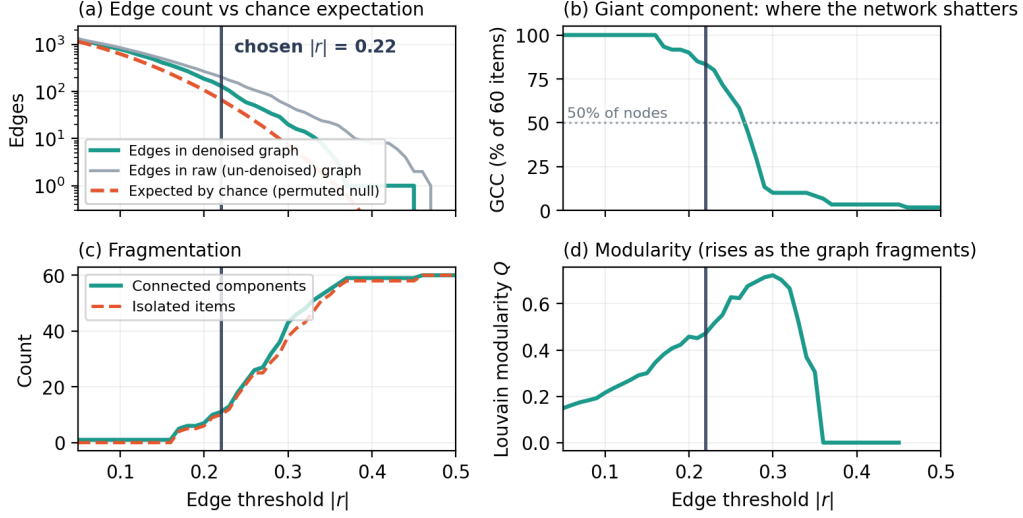


Figure 8: The signed item network rebuilt at all 46 cutoffs from 0.05 to 0.50. (a) Edge count against the chance expectation from Graph 4. (b) Giant component. (c) Components and isolates. (d) Louvain modularity, which reaches 0.72 only once the graph has broken into 43 pieces. The vertical line marks the selected $|r| > 0.22$.

and written examinations — while endorsing curriculum reform above 94%. Within technology the objection tracks delegated consequential judgment (diagnosis, autonomous vehicles) rather than automation as such: robots in hazardous work still score 82.4%.

Most apparent structure was response style. Row-centring removes 87% of the item pairs exceeding $|r| \geq 0.30$. Any analysis skipping this step finds dense, uniformly positive structure and is measuring agreeableness.

Opinions are internally consistent. Structural balance (Figure 12) rises monotonically to 100% of triads as weak edges are removed, against a sign-permutation null flat at 50%. A spurious effect would not sharpen as noise is filtered out.

Communities ignore the survey’s own design. The 15 detected communities align poorly with the four thematic blocks ($\text{NMI} = 0.205$). Respondents associate ideas across topic boundaries, not within the categories the questionnaire was built from.

4.2 What failed, and what that tells us

Three results did not come out the way the analysis was set up to expect. They are reported because each one changes what may legitimately be claimed.

1. Most of the correlation matrix is noise. Marchenko–Pastur filtering (Section 3.5) leaves exactly four eigenvalue modes above the band, carrying 25.6% of total variance. The other 74.4% is what 91 respondents answering 60 items generate by chance. Any claim resting on a single correlation, or on fine structure of the full matrix, is unsupported — which is why every downstream network is built from the denoised matrix rather than the raw one.

2. The community structure does not survive its own null. This is the sharpest negative result, and which null is used decides it. A correlation-derived network has transitivity built into its construction: if A correlates with B and B with C, A and C are pulled toward correlation automatically, and that alone manufactures modularity in structureless data. Rewiring [6] and Erdős–Rényi nulls take the finished graph as given, so neither can detect structure the construction itself created. Only a column-permutation null — destroying association in the raw responses, then regenerating the entire chain of centre, correlate, threshold, measure — tests the claim being made. Both networks were tested independently and both return the same verdict:

Graph 9: topic-block connectivity, 91 respondents x 60 items

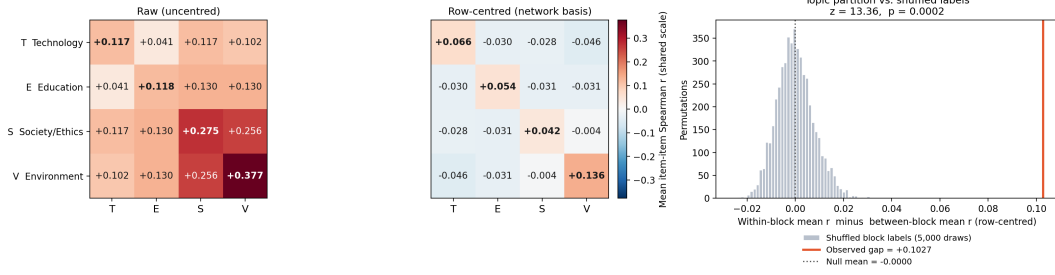


Figure 9: Topic-block connectivity. Left: mean item-item correlation within and between blocks on the raw matrix — every cell positive, the design invisible under acquiescence. Centre: the same collapse after row-centring, shared colour scale; diagonal positive, all off-diagonals negative. Right: null distribution of the within-minus-between gap under 5,000 reshufflings of the block labels. The observed gap +0.1027 ($z = 13.4$) is not approached by any permutation.

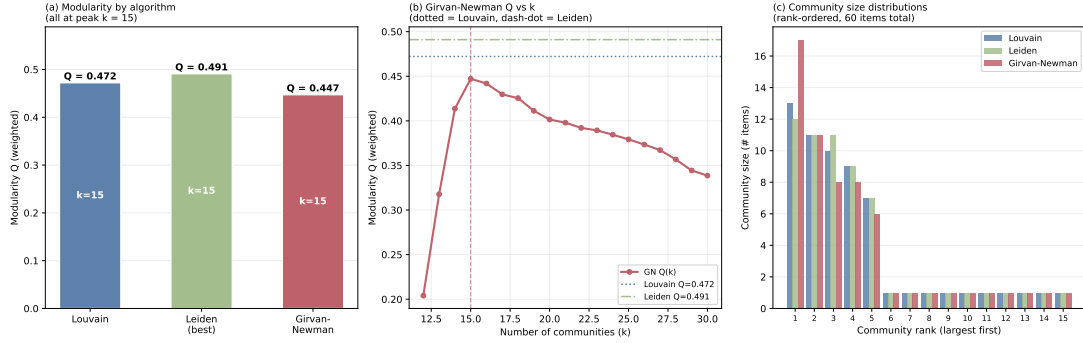


Figure 10: Three-algorithm comparison on the item network. (a) Peak modularity per algorithm; all converge on $k = 15$, Leiden highest at $Q = 0.491$. (b) The Girvan-Newman $Q(k)$ curve rising to a clear peak at $k = 15$ before declining, with Louvain and Leiden peaks marked. (c) Rank-ordered community sizes, similar across all three methods.

Network	Null	Q_{null}	z
Respondent ($Q = 0.421$)	Rewiring	$0.337 \pm .011$	7.46
	Erdős-Rényi	$0.337 \pm .011$	7.28
	Permutation	$0.404 \pm .018$	0.91
Item ($Q = 0.491$)	Rewiring	$0.333 \pm .013$	12.13
	Erdős-Rényi	$0.420 \pm .016$	4.46
	Permutation	$0.629 \pm .044$	-3.14

Against the graph-level nulls the structure looks overwhelming, at z up to 12. Against the null that reproduces the construction it vanishes on the respondent network ($p = 0.19$) and inverts on the item network: scrambled data pushed through the identical pipeline yields more modular graphs than the real data. Had we stopped at the conventional rewiring null we would have reported a strong, entirely spurious finding. The communities are real as a description of the graph; they are not evidence about the cohort.

3. The dimension count depends on who you keep. Filling the same 242 cells by column mean, column me-

dian or ordinal regression all yield exactly four modes (24.7%, 25.0%, 25.6% of variance) — the structure is a property of the data, not of the imputer. But dropping the ten respondents with more than twelve missing items and filling the rest with a column statistic gives only three modes at 19.3%. The count is robust to how missing cells are filled and fragile to which respondents are retained. Keeping all 91 is the better-supported choice — discarding usable respondents to avoid modelling 4.4% of cells trades away more than it protects — but the result is conditional on it.

4.3 Methods tried and rejected

Alternatives were measured rather than argued about.

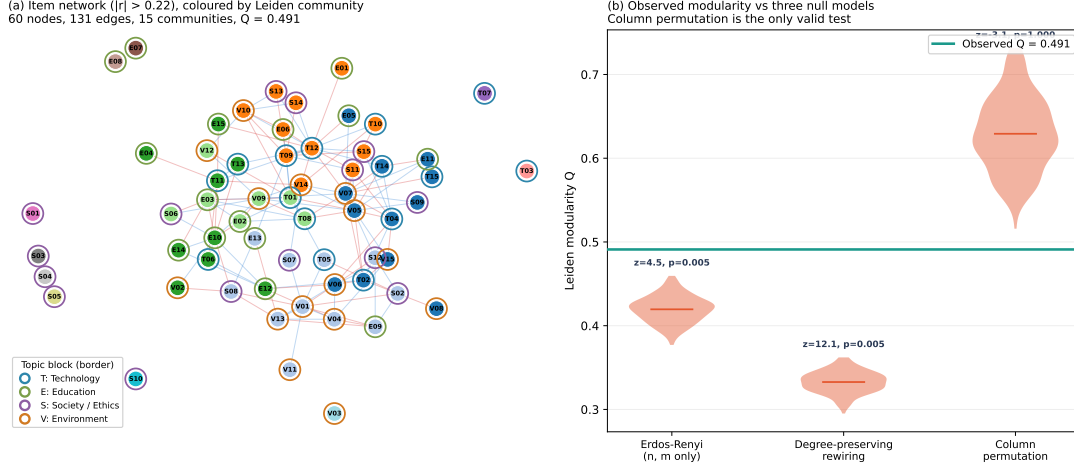


Figure 11: (a) The item network at $|r| > 0.22$, nodes coloured by Leiden community, borders marking the predefined thematic block. (b) Observed modularity $Q = 0.491$ sits comfortably above the Erdős-Rényi and degree-preserving rewiring nulls but below the column-permutation null, mirroring the artefact found independently in the respondent network.

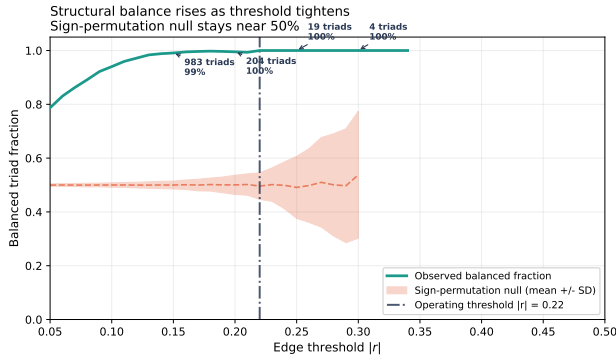


Figure 12: Structural balance as a function of the correlation threshold $|r|$. The fraction of balanced triads rises towards 100% as the threshold tightens and spurious edges are removed, while the sign-permutation null distribution remains anchored near 50%. The operating threshold of $|r| = 0.22$ is marked.

4.4 Limitations

Items with means above roughly 4.5 measure social desirability more than preference strength, and with few reverse-worded items acquiescence and a genuine ceiling cannot be fully separated. The sample is $n = 96$ from a single institution, so only large gaps between statements sit safely above sampling variation, and monotone dropout leaves later blocks on a slightly smaller base. No demographic fields exist, so nothing can be attributed to a subgroup. Finally, and most importantly for the network claims: correlation stability does not imply partition stability. Across NC treatments the correlation matrices agree at $r > 0.94$ while their community assignments agree at AMI as low as 0.337. Community-level conclusions need their own bootstrap stability analysis before being treated as findings.

5 Conclusion

We built an end-to-end pipeline from a raw ordinal questionnaire to a signed opinion network and a process on it, measuring each intervention rather than asserting it: ordinal-regression recovery of 242 cells beats a modal baseline under both masking mechanisms, and row-centring eliminates 87% of item pairs correlated at $|r| \geq 0.30$. A permutation null puts the noise floor at 0.106 against an observed spread of 0.139, Marchenko–Pastur filtering leaves four modes carrying 25.6% of variance, and the threshold came from a 46-point sweep by a stated rule at $|r| > 0.22$.

The same framework produced the most useful negative result. Community structure clears rewiring and

Approach	Why it was rejected
Recode NC as Neutral	Imposes a 0.92-point downward shift at those cells; asserts a response that was explicitly withheld.
Drop the 17 NC users	Discards a fifth of the sample to remove 37 cells, and is the only NC treatment that disturbs the recovered partition (AMI 0.337).
Mutual k NN (respondents)	Shatters the graph: 66 edges, 37 components, giant component 49.5%. Union k NN keeps one component.
Raw uncentred similarity	624 of 1,770 pairs survive $ r > 0.22$, 83% of them positive — a 35%-dense graph measuring agreeableness.
Percentile threshold (p90)	Fixes the edge count by construction, so it cannot say whether any edge beats chance. On respondents it leaves 22 components and 20 isolates.
Louvain for communities	Superseded by Leiden: its greedy phase can leave communities internally disconnected.

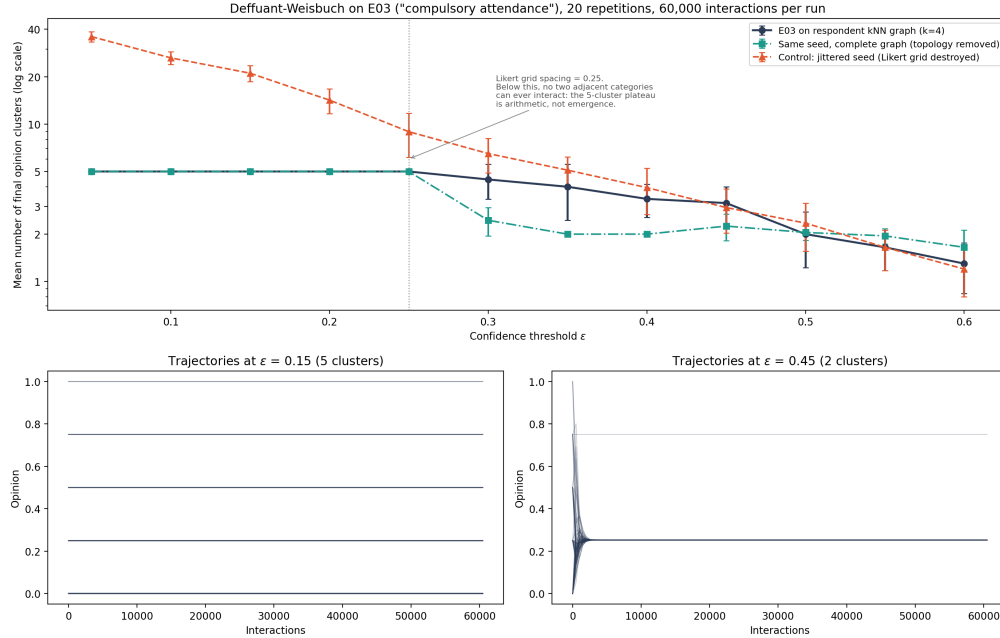


Figure 13: Deffuant–Weisbuch dynamics seeded on E03. Left: final cluster count against confidence threshold ϵ for the k NN respondent network, a complete graph, and a jittered-seed control. The flat 5.00 ± 0.00 plateau below $\epsilon = 0.25$ is forced by the five-point grid and vanishes under jitter. Right: above the plateau the sparse network sustains more opinion clusters than the well-mixed population at the same tolerance.

Erdős–Rényi nulls on both networks and fails the permutation null on both — a finding we would have missed entirely had we used only the conventional null, and the clearest argument in this report for choosing a null that reproduces the construction. What survives is a cohort in superficial consensus with genuine but local structure: four modest cross-cutting dimensions, communities that ignore the survey’s own thematic design (NMI 0.205), and a signed topology that becomes perfectly balanced as noise is stripped away. Not two camps.

Individual Contribution

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Table 2: Contribution by team member.

Member	Contribution
Arijeet Paul	Data sanitation and missingness analysis; ordinal-regression imputation and its masking validation; row-centring; Graphs 1, 2, 3 and 12; respondent k NN network; repository structure and run_all.py.
Shreyash Lohare	Noise-floor analysis and the permutation null framework (src/common/nulls.py); Marchenko–Pastur spectral denoising and the denoised correlation matrix; the threshold sweep and its selection rule; the three-null comparison on the respondent network (Section 4.2); sensitivity analysis of the rejected methods (Section 4.3); shared configuration and analysis matrix (src/common/, src/analysis/); Graphs 4, 5 and 7; repository structure and run_all.py.
Dev Patel	Community detection on the item network and the Louvain / Leiden / Girvan–Newman comparison; null-model evaluation of the item partition (Graph 10); structural balance sweep against a sign-permutation null (Graph 11); the reordered MP-denoised correlation heatmap (Graph 6); src/communities/; report sections for Graphs 6, 10 and 11.

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